

# Avinash Tiwari, Ph.D. Student

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✉ itsmeavinash@gmail.com

✉ avinash.tiwari@iucaa.in

🌐 <https://avinash-tiwari-at.github.io/>

🆔 0000-0001-7197-8899

## Current Affiliations

- 2021 – . . . . . 📌 Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune.
- 2022 – . . . . . 📌 LIGO Scientific Collaboration (LSC) and LIGO-India Scientific Collaboration (LISC).

## Education

- 2021 – . . . . . 📌 **Ph.D.**, Inter-University Centre for Astronomy and Astrophysics, Pune.
- 2019 – 2021 📌 **M.Sc. Physics**, Indian Institute of Technology Guwahati, Guwahati.
- 2016 – 2019 📌 **B.Sc. Physics, Mathematics, & Computer Science**, University of Allahabad, Prayagraj.

## Research Background

- 📌 **Data Analysis:** Parameter estimation, Fisher matrix analysis, and hierarchical inference. **Waveform Modeling:** Analytical modeling of line-of-sight acceleration and velocity waveforms. **Probing Merger Environments:** developing methods for probing CBC environments on a single-event basis. **Characterizing GW signals:** Developing model-independent tests to characterize the GW signals. **Cosmology:** Early universe cosmology driven by the synergies between GW and 21-cm global signal observations.

## Skills

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|-------------------------------|----------------------------------------------------------------|
| Languages                     | 📌 English, Hindi.                                              |
| Programming                   | 📌 Python, C, Mathematica.                                      |
| Bayesian Inference & Sampling | 📌 BILBY, BILBY_PIPE, BILBY_TGR, GWPOPULATION, EMCEE, DYNESTY.  |
| Open Source Contributions     | 📌 LOSA pipeline within the BILBY_TGR package (Core developer). |

## Awards & Honors

- 📌 **Junior Research Fellowship (JRF):** Awarded by the Council of Scientific and Industrial Research (CSIR), Government of India (All India Rank 89 in Physical Sciences). (2020).
- 📌 **Institute Merit-cum-Means Scholarship:** Awarded by the Indian Institute of Technology Guwahati for academic excellence (top 25% of the M.Sc. cohort). (2019 – 2021).

## Conferences, Workshops, and Colloquia

- 📌 **GW - Detector Characterization Workshop** (December 2025), IUCAA, Pune.
- 📌 **GR24 - Amaldi16** (July 2025). Talk: “Profiling stellar environments of gravitational wave sources”.
- 📌 **AEI Colloquium** (August 2024), AEI Hanover: “Profiling stellar environments of gravitational wave sources”.

## Conferences, Workshops, and Colloquia (continued)

- **New ideas on the ORIGIN of BLACK HOLE MERGERS** (August 2024), NBI Copenhagen. **Invited Talk:** “Profiling stellar environments of gravitational wave sources.”
- **ICGC-2023** (December 2023). Talk: “Probing globular clusters and other astrophysical environments with gravitational waves emitted by accelerated compact binary mergers.”
- **Amaldi15** (July 2023). Talk: “Probing globular clusters and other astrophysical environments with gravitational waves emitted by accelerated compact binary mergers.”
- **Discussion Meeting on LUNAR GRAVITATIONAL-WAVE DETECTION** (April 17 - 20, 2023), Ramanujan Lecture Hall, ICTS Bengaluru.
- **IAGRG32** (December 2022). Poster presentation and a Flash Talk: “Waltzing binaries: probing line-of-sight acceleration of merging compact objects with gravitational waves”.

## Other Talks / Seminars

- **Last Friday Talks at IUCAA** (August 29, 2025): “Profiling Dark Matter Spikes with Gravitational Waves from Accelerated Binaries”.
- **Last Friday Talks at IUCAA** (September 27, 2024): “Profiling stellar environments of gravitational wave sources”.
- **Last Friday Talks at IUCAA** (July 28, 2023): “Probing globular clusters and other astrophysical environments with gravitational waves emitted by accelerated compact binary mergers”.
- **Last Friday Talks at IUCAA** (March 31, 2023): “Waltzing binaries: probing line-of-sight acceleration of merging compact objects with gravitational waves”.
- **Online Seminar at ICTS, Bengaluru** (December 15, 2022): “Waltzing binaries: probing line-of-sight acceleration of merging compact objects with gravitational waves”.

## Courses Tutored

- PHY-IC481 - General Relativity at IUCAA.
- PHY-IE485 - Elective course - Cosmology at IUCAA.

## Mentoring

- Sudhir Gholap (IUCAA)
- Md Asim Ansari (IUCAA)

## Journal Referee

- Monthly Notices of the Royal Astronomical Society

## References

- **Prof. Shasvath J. Kapadia**, IUCAA, shasvath.j.kapadia@gmail.com
- **Dr. Aditya Vijaykumar**, CITA, aditya@utoronto.ca
- **Prof. Sourav Chatterjee**, TIFR, souravchatterjee.tifr@gmail.com
- **Prof. Basudeb Dasgupta**, TIFR, bdasgupta@theory.tifr.res.in
- **Prof. Susmita Adhikari**, IISER Pune, susmita@iiserpune.ac.in
- **Prof. Tirthankar Roy Choudhury**, NCRA-TIFR, troych75@gmail.com